Let $S = \frac{1}{2}$. Then $U = e^{i\frac{\pi}{2}\sigma^x}$ $U^+ = e^{-i\frac{\pi}{2}\sigma^x}$

$$U \sigma^x U^+ = \sigma^x$$

$$U \sigma^y U^+ = \left(\cos\frac{\pi}{2} + i\sigma^x \sin\frac{\pi}{2}\right) \sigma^y \left(\cos\frac{\pi}{2} - i\sigma^x \sin\frac{\pi}{2}\right) =$$

$$= i\,\sigma^x \sigma^y (-i\sigma^x) = -i\,\sigma^x\sigma^z = (-i)^2\sigma^y = -\sigma^y$$

$$\sigma^y\sigma^x = -i\sigma^z$$

$$\sigma^x\sigma^z = i\,\varepsilon^{xzy}\,\sigma^y = -i\sigma^y$$

$$U\sigma^z U^+ = i\sigma^x\sigma^z(-i\sigma^x) = i\sigma^x\sigma^y = i^2\sigma^z = -\sigma^z$$

$$\sigma^z\sigma^x = i\,\varepsilon^{zxy}\,\sigma^y = i\sigma^y$$

Thus operator $U = e^{i\frac{\pi}{2}\sigma^x}$ describes rotation about axis (σ^x) by an angle (π) so that

$$\vec{S} = \frac{\hbar}{2}\vec{\sigma} = \frac{\hbar}{2}\begin{pmatrix}\sigma^x\\ \sigma^y\\ \sigma^z\end{pmatrix} \longrightarrow \frac{\hbar}{2}\begin{pmatrix}\sigma^x\\ -\sigma^y\\ -\sigma^z\end{pmatrix} = \begin{pmatrix}S^x\\ -S^y\\ -S^z\end{pmatrix}.$$

$$H_{AFM} = J \sum_{\substack{n\in A\\ g}} \vec{S}_n \vec{S}_{n+g}$$

The idea is to rotate every spin on B-sublattice

$$H \to U H U^+ \quad \text{where} \quad U = \prod_{m\in B} e^{i\frac{\pi}{2}\sigma^x_m} = \prod_{m\in B}\left(i\sigma^x_m\right)$$

$$U H U^+ = \frac{1}{2}J \sum_{\substack{n\\ g}} S^z_n\left(-S^z_{n+g}\right) + S^x_n\left(S^x_{n+g}\right) + S^y_n\left(-S^y_{n+g}\right) =$$

$$= -\frac{1}{2}J \sum_{\substack{n\\ g}} \left(S^z_n S^z_{n+g} + S^y_n S^y_{n+g} - S^x_n S^x_{n+g}\right)$$

<u>Note</u>: these are rotated spins!

$$S_n^y S_{n+g}^y - S_n^x S_{n+g}^x = \frac{1}{(2i)^2}\left(S_n^+ - S_n^-\right)\left(S_{n+g}^+ - S_{n+g}^-\right)$$

$$- \frac{1}{2^2}\left(S_n^+ + S_n^-\right)\left(S_{n+g}^+ + S_{n+g}^-\right) = -\frac{1}{4}\Big(\underline{S_n^+ S_{n+g}^+} - \cancel{S_n^+ S_{n+g}^-}$$

$$- \cancel{S_n^- S_{n+g}^+} + \underline{\underline{S_n^- S_{n+g}^-}} + \underline{S_n^+ S_{n+g}^+} + \cancel{S_n^+ S_{n+g}^-} + \cancel{S_n^- S_{n+g}^+} + \underline{\underline{S_n^- S_{n+g}^-}}\Big)$$

$$= -\frac{1}{2}\left(S_n^+ S_{n+g}^+ + S_n^- S_{n+g}^-\right)$$

$$U H U^+ = \frac{1}{2} J \sum_{n\, g} \left\{ \frac{1}{2}\left(S_n^+ S_{n+g}^+ + S_n^- S_{n+g}^-\right) - S_n^z S_{n+g}^z \right\}$$

But the point is that now we do not need to introduce two types of bosons — our rotated basis has become coordinate-dependent, and only <u>one</u> boson is needed.

$$S_n^z = S - a_n^+ a_n \quad , \quad S_n^+ \cong \sqrt{2S}\, a_n , \quad S_n^- \cong \sqrt{2S}\, a_n^+$$

$$\tilde{H} = U H U^+ = \frac{1}{2} J \sum_{n\, g} \Big\{ -S^2 + S\left(a_n^+ a_n + a_{n+g}^+ a_{n+g}\right) + \frac{1}{2}(2S)\left(a_n a_{n+g} + a_n^+ a_{n+g}^+\right) \Big\} = \tilde{H}^{(0)} + \tilde{H}^{(2)}$$

$$\tilde{H}^{(2)} = \frac{1}{2} J S \sum_{n\, g} \left(a_n^+ a_n + a_{n+g}^+ a_{n+g} + a_n^+ a_{n+g}^+ + a_n a_{n+g}\right)$$

$$\tilde{H}^{(0)} = -\frac{1}{2} J S^2 \sum_{n\, g} 1 = -\frac{1}{2} J S^2 \cdot (Nz) \text{ (as before)}$$

$$a_n = \frac{1}{\sqrt{N}} \underbrace{\sum_{k \in BZ}}_{\text{full BZ of the lattice!}} e^{-i\vec{k}\vec{R}_n}\, a_k$$

$$\Rightarrow \tilde{H}^{(2)} = \frac{1}{2} J S \sum_g \sum_k \left\{ 2 a_k^+ a_k + a_k^+ a_{-k}^+ e^{i\vec{k}\vec{g}} + a_k a_{-k} e^{-i\vec{k}\vec{g}} \right\}$$

$$= \frac{1}{2} JS \sum_k \overbrace{z(a_k^+ a_k + a_{-k}^+ a_{-k})}^{2a_k^+ a_k} + z\gamma_k (a_k^+ a_{-k}^+ + a_k a_{-k})$$

where $\gamma_k \equiv \frac{1}{z} \sum_g e^{i\vec{k}\vec{g}} = \frac{2}{z}(\cos k_x a + \cos k_y a + \cos k_z a)$

($z = 2d$ for hypercubic lattice) $= \frac{1}{d} \sum_{i=1}^{d} \cos k_i a$.

Bogoliubov transformation again (in fact, it was used earlier by Holstein, Primakoff in 1940!)

$$\left.\begin{aligned} a_k &= u_k \alpha_k - v_k \alpha_{-k}^+ \\ a_k^+ &= u_k \alpha_k^+ - v_k \alpha_{-k} \end{aligned}\right\}$$

transformation very similar to previous case with two bosons — (a_{-k}) plays the role of the "other mode".

We chose u_k, v_k be real and symmetric $u_k = u_{-k}$, $v_k = v_{-k}$

$$\hat{H}^{(2)} = \frac{zJS}{2} \sum_k \Big\{ (u_k \alpha_k^+ - v_k \alpha_{-k})(u_k \alpha_k - v_k \alpha_{-k}^+) +$$

$$+ (u_k \alpha_{-k}^+ - v_k \alpha_k)(u_k \alpha_{-k} - v_k \alpha_k^+) + \gamma_k \Big[(u_k \alpha_k^+ - v_k \alpha_{-k}) \cdot$$

$$\cdot (u_k \alpha_{-k}^+ - v_k \alpha_k) + (u_k \alpha_{+k} - v_k \alpha_{-k}^+) \cdot (u_k \alpha_{-k} - v_k \alpha_k^+) \Big] =$$

$$= \frac{1}{2} zJS \sum_k \Big\{ u_k^2 (\alpha_k^+ \alpha_k + \underbrace{\alpha_{-k}^+ \alpha_{-k}}_{\to 1 + \alpha_k^+ \alpha_k}) + v_k^2 (\alpha_{-k} \alpha_{-k}^+ + \alpha_k \alpha_k^+)$$

$$- \gamma_k \Big[u_k v_k (\alpha_k^+ \alpha_k + \alpha_{-k} \alpha_{-k}^+) + u_k v_k (\alpha_k \alpha_k^+ + \alpha_{-k}^+ \alpha_{-k}) \Big]$$

$$+ \alpha_k^+ \alpha_{-k}^+ (-2u_k v_k + \gamma_k (u_k^2 + v_k^2)) + \alpha_{-k} \alpha_k (-2u_k v_k + \gamma_k (u_k^2 + v_k^2)) \Big\}$$

$$\hat{H}^{(2)} = \frac{zJS}{2} \sum_k (v_k^2 - 2\gamma_k u_k v_k) + \frac{1}{2} JSz \sum_k (u_k^2 + v_k^2 - 2\gamma_k u_k v_k) \cdot$$

$$\cdot (\alpha_k^+ \alpha_k + \alpha_{-k}^+ \alpha_{-k}) .$$

and we require <u>no</u> anomalous terms: (*) $\gamma_k (u_k^2 + v_k^2) = 2u_k v_k$

We also check commutator:

$$[a_k, a_k^+] = [u_k \alpha_k - v_k \alpha_{-k}^+, u_k \alpha_k^+ - v_k \alpha_{-k}] = u_k^2 [\alpha_k, \alpha_k^+] + v_k^2 [\alpha_{-k}^+, \alpha_{-k}]$$

$$= u_k^2 - v_k^2 = 1$$

$\Rightarrow$ thus as before can choose $\begin{cases} u_k = \cosh \varphi_k \\ v_k = \sinh \varphi_k \end{cases}$

Eq (*) : $\gamma_k \cosh 2\varphi_k = \sinh 2\varphi_k \Rightarrow \tanh 2\varphi_k = \gamma_k$.

Spectrum is proportional to the combination :

$$u_k^2 + v_k^2 - 2\gamma_k u_k v_k = \cosh 2\varphi_k - \gamma_k \sinh 2\varphi_k =$$

$$= \underbrace{\cosh 2\varphi_k}_{= \frac{1}{\sqrt{1-\tanh^2(2\varphi_k)}}} (1-\gamma_k^2) = \sqrt{1-\gamma_k^2}$$

Thus $\tilde{H}^{(2)} = JSz \sum_k \sqrt{1-\gamma_k^2}\, \alpha_k^\dagger \alpha_k$ + constant

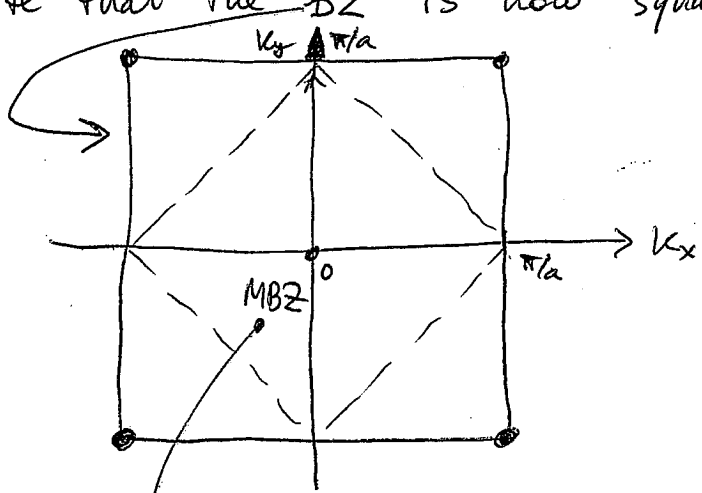
$\parallel$

correction to ground state energy

The spectrum of spin waves (α-bosons)

$$\hbar \omega_k = JSz\sqrt{1-\gamma_k^2}$$

Note that the BZ is now square $\left(-\frac{\pi}{a}, \frac{\pi}{a}\right) \times \left(-\frac{\pi}{a}, \frac{\pi}{a}\right)$



and includes point $\left(\frac{\pi}{a}, \frac{\pi}{a}\right)$ (and other 3 points, corners of BZ, that are equivalent to it)

$\gamma_k = 1$ at $\vec{k} = \vec{0}$ (as before)

AND $\vec{k} = \left(\frac{\pi}{a}, \frac{\pi}{a}, \frac{\pi}{a}\right)$

antiferromagnetic vector

previously, by introducing <u>two</u> sublattices (A and B) we had to work in the <u>magnetic BZ</u> which does not include $\left(\frac{\pi}{a}, \frac{\pi}{a}, \frac{\pi}{a}\right)$ point!

Spin reduction

$$\langle S_n^z \rangle = S - \langle a_n^+ a_n \rangle = S - \frac{1}{N} \sum_{k_1 k_2} \langle a_{k_1}^+ a_{k_2} \rangle e^{i(\bar{k}_1 - \bar{k}_2)\cdot \bar{R}_n}$$

$$= S - \frac{1}{N} \sum_{k_1 k_2} e^{i(\bar{k}_1 - \bar{k}_2)\bar{R}_n} \underbrace{\langle (u_{k_1} \alpha_{k_1}^+ - v_{k_1} \alpha_{-k_1})(u_{k_2} \alpha_{k_2} - v_{k_2} \alpha_{-k_2}^+) \rangle}_{u_{k_1} u_{k_2} \underbrace{\langle \alpha_{k_1}^+ \alpha_{k_2} \rangle}_{\delta_{k_1,k_2} \langle \alpha_{k_1}^+ \alpha_{k_1} \rangle} + v_{k_1} v_{k_2} \underbrace{\langle \alpha_{-k_1} \alpha_{-k_2}^+ \rangle}_{\delta_{k_1,k_2} 1 + \langle \alpha_{-k_2}^+ \alpha_{-k_1} \rangle}}$$

$$= S - \frac{1}{N} \sum_k \left[u_k^2 \langle \alpha_k^+ \alpha_k \rangle + v_k^2 \left(1 + \langle \alpha_{-k}^+ \alpha_{-k} \rangle\right) \right]$$

$$= S - \frac{1}{N} \sum_k \left\{ \underbrace{(u_k^2 + v_k^2)}_{ch^2\varphi_k + sh^2\varphi_k = ch 2\varphi_k} \langle \alpha_k^+ \alpha_k \rangle + \underbrace{v_k^2}_{sh^2\varphi_k = \frac{1}{2}(ch 2\varphi_k - 1)} \right\}.$$

Recall $ch 2\varphi_k = \dfrac{1}{\sqrt{1 - \tanh^2(2\varphi_k)}} = \dfrac{1}{\sqrt{1 - \gamma_k^2}}$

$$\langle S_n^z \rangle = S - \frac{1}{N} \sum_k \left\{ \frac{1}{\sqrt{1-\gamma_k^2}} \underbrace{\langle \alpha_k^+ \alpha_k \rangle}_{\frac{1}{e^{\frac{\hbar\omega_k}{k_B T}} - 1}} + \underbrace{\frac{1}{2}\left(\frac{1}{\sqrt{1-\gamma_k^2}} - 1\right)}_{T=0 \text{ contribution (due to quantum fluctuations)}} \right\}$$

(finite at $T>0$)

$$\langle S_n^z \rangle = S - \Delta S_T - \Delta S_0$$

$$\Delta S_0 = \frac{1}{N} \sum_k \frac{1}{2}\left(\frac{1}{\sqrt{1-\gamma_k^2}} - 1\right) = \frac{1}{2} \cdot \frac{1}{(2\pi)^2} \iint_{BZ} \frac{dx\,dy}{\sqrt{1-\gamma_k^2}}$$

$d=2$

Zero-temperature quantum fluctuations

$$\Delta S_T = \frac{1}{N}\sum_k \frac{1}{\sqrt{1-\gamma_k^2}}\,\frac{1}{e^{\frac{\hbar\omega}{k_BT}}-1} = \frac{1}{(2\pi)^2}\iint dx\,dy\,\frac{1}{\sqrt{1-\gamma_k^2}}\,\frac{1}{e^{\frac{\hbar\omega_k}{k_BT}}-1}$$

For small k, which is most important at low $T \geq 0$:

$$\hbar\omega_k \approx JS\underset{\substack{\parallel\\ 2d\\ \parallel\\ 4}}{z}\sqrt{1-\frac{1}{4}\left(1-\frac{1}{2}x^2+1-\frac{1}{2}y^2\right)} =$$

$$= 4JS\sqrt{\frac{x^2+y^2}{8}} = JSa\sqrt{2}\,|k|.$$

$$\Rightarrow \Delta S_T \cong \frac{1\,2}{(2\pi)^2}\int dx\,dy\,\frac{\sqrt{8}}{\sqrt{x^2+y^2}}\,\frac{k_BT}{JS\sqrt{2}\sqrt{x^2+y^2}} =$$

contribution from $|\vec{k}| \sim 0$ and $\left(\frac{\pi}{a}, \frac{\pi}{a}, \frac{\pi}{a}\right)$!

$$= \frac{1}{\pi^2}\,\frac{k_BT}{JS}\int_0^{\Lambda\sim\frac{\pi}{a}}\frac{dx\,dy}{x^2+y^2} = \frac{2k_BT}{\pi SJ}\int_0^{\Lambda}\frac{dr\cdot r}{r^2} =$$

$$\left(\text{imagine lower cut-off } k_0\right) = \frac{2k_BT}{\pi SJ}\ln\frac{\Lambda}{k_0} \xrightarrow[k_0\to 0]{} \infty$$

Arbitrary small temperature destroys magnetic order in $d=2$ Antiferromagnet!

<u>Things are even worse in $d=1$ AFM</u>

$$\Delta S_T = \frac{1}{2\pi}\int dx\,\frac{k_BT}{2JS\cdot\sin^2 x} = \frac{2k_BT}{4\pi JS}\int_0^{\Lambda}\frac{dx}{x^2} = \frac{k_BT}{2\pi JS}\left(\frac{1}{k_0}-\frac{1}{\Lambda}\right)$$

$k_0 \to 0$

$\sqrt{1-\gamma_k^2} = \sqrt{1-\cos^2 x} = |\sin x|$

$\hbar\omega_k = 2JS\,|\sin x|$

The real problem is of course: $\Delta S_0 \sim \int \frac{dx}{|\sin x|} \sim \ln 0 = \infty$ $\boxed{T=0 \text{ correction diverges!}}$

$\underline{d=3}$ No problem as $\hbar\omega_k \sim k$ but the integration measure is $k^2 dk \Rightarrow$

$$\Delta S_T \sim \int dk \cdot k^2 \cdot \frac{1}{k} \frac{k_B T}{JSk} \sim O(k_B T)$$

(no singularity in the integral)

Let us estimate the <u>correlation length</u> ξ (length on which the magnetic order is lost at finite $T \neq 0$):

$\Delta S_T \sim S$ meaning the reduction is complete

$$\Delta S_T = \frac{2 k_B T}{\pi J S} \ln \frac{\Lambda}{k_0}$$

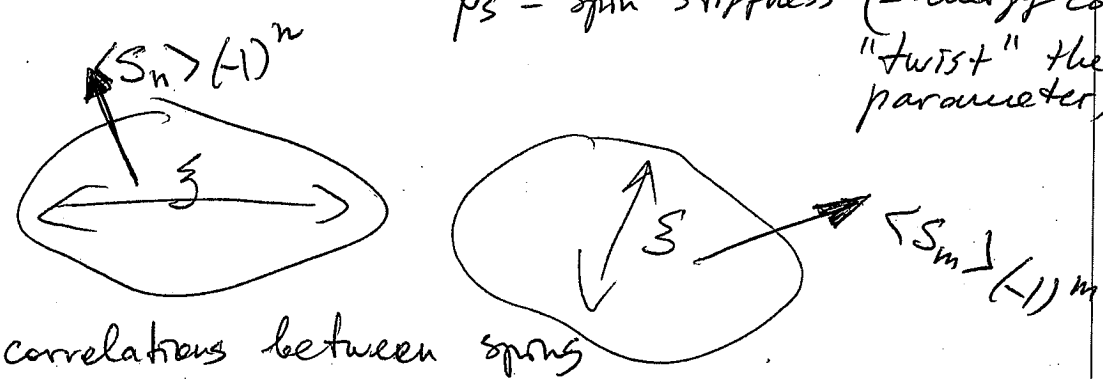
$$\Rightarrow \ln \frac{\Lambda}{k_0} = c \frac{JS^2}{k_B T} \quad (c = \text{number})$$

$$\Rightarrow \frac{\Lambda}{k_0} = \exp\left(\frac{c S^2 J}{k_B T}\right)$$

Associated spatial scale $\xi \sim k_0^{-1} = \Lambda^{-1} \exp\left(+\frac{c S^2 J}{k_B T}\right)$

Exact result $\xi = a \exp\left(+\frac{\rho_s}{k_B T}\right)$ exponentially large!

ρ_s = spin stiffness (= energy cost to "twist" the order parameter)



No correlations between spins at distances above ξ.

Is this specific to $d=2$ AFM? No!

Check out previously studied ferromagnet (FM):

$$\langle S_n \rangle = S - \frac{1}{N}\sum_k n_k = S - \frac{1}{(2\pi)^d}\int d^d\vec{k}\,\frac{1}{e^{\frac{\hbar\omega_k}{k_BT}}-1}$$

where $\hbar\omega_k \sim k^2$ for $k \to 0$

Then $d=2$

$$\Delta S_T = \frac{1}{(2\pi)^2}\frac{k_BT}{D}\int_0^\Lambda \frac{dk\cdot k\cdot 2\pi}{k^2} =$$

$$= \frac{k_BT}{2\pi D}\ln\frac{\Lambda}{k_0} \xrightarrow[k_0\to 0]{} \infty$$

(the same log-divergence!)

$d=1$

$$\Delta S_T \cong \frac{1}{2\pi}\int_{k_0}^{\Lambda} dk\,\frac{k_BT}{Dk^2} \approx \frac{k_BT}{\pi D}\left(\frac{1}{k_0}-\cancel{\frac{1}{\Lambda}}\right)$$

$$\Rightarrow \quad \frac{k_BT}{D}\frac{1}{k_0} \sim S \Rightarrow \frac{1}{k_0} \sim \frac{DS}{k_BT} \sim \frac{S^2J}{k_BT}$$

$\Rightarrow$ spin correlation length in $d=1$ goes as $\xi \sim \frac{S^2J}{k_BT}$.

All of this is a consequence of Mermin–Wagner theorem:

No magnetic long range order (that is, $\langle S_R \rangle = 0$) for magnetic systems with <u>continuous</u> rotational symmetry in dimensions 1 and 2.

	$d=1$	$d=2$	$d=3$
Heisenberg	NO	NO	YES
Ising	NO	YES	YES
XY	NO	"kind of" BKT	YES

$\boxed{T>0}$